「Rationalize.

Let K be a field. Given $\theta \in K$, find an irreducible polynomial

$$P(x) = p_n x^n + p_{n-1}x^{n-1} + \cdots + p_1 x + p_0 \quad \text{s.t.} \quad P(\theta) = 0.$$

Then, $\theta^{-1} = -p_0^{-1} \cdot (p_n \cdot \theta^{n-1} + p_{n-1}\theta^{n-2} + \cdots + p_1) \in K$ is the multiplicative inverse of θ.

「Problem. Rationalize the denominator of $\dfrac{1}{3 + 2\cdot\sqrt[3]{2} + \sqrt[3]{4}}$.

Proof. In other words, find $(3 + 2\cdot\sqrt[3]{2} + \sqrt[3]{4})^{-1}$ as $\mathbb{Q}$-linear combination for $\sqrt[3]{2}$.

Let's calculate. By the Euclidean Algorithm,

$$x^3 - 2 = (x^2 + 2x + 3)\cdot(x - 2) + (x + 4)$$
$$x^2 + 2x + 3 = (x + 4)(x - 2) + 11$$
$$x + 4 = 11 \cdot \left(\tfrac{1}{11}x + \tfrac{4}{11}\right) \quad \text{(Not necessary term)}$$

So, $(x + 4) = (x^3 - 2) - (x^2 + 2x + 3)\cdot(x - 2)$

Put

$(x + 4)(x - 2) + 11 = x^2 + 2x + 3$

Then, we obtain $(x - 2)\cdot(x^3 - 2) - (x^2 + 2x + 3)\cdot(x^2 - 4x + 5) = -11$.

Since $\theta = \sqrt[3]{2}$ satisfies that $\theta^3 - 2 = 0$, so

$$(\theta^2 + 2\theta + 3)\cdot(\theta^2 - 4\theta + 5) = 11, \text{ or}$$

$$(3 + 2\cdot\sqrt[3]{2} + \sqrt[3]{4})^{-1} = \frac{1}{11}\cdot(\sqrt[3]{4} - 4\cdot\sqrt[3]{2} + 5). \quad \blacksquare$$

「In general, let $P(x)$ be an irreducible polynomial over F.

Then, $K = F[x]/(P(x))$ has a root of $P(x)$, namely α.

For F-polynomial $f(x)$, the inverse of $f(\alpha) \neq 0$ is:

$(f(x)) + (P(x)) = (1)$, since $F[x]$ is P.I.D and $P(x)$ is irreducible,

So, $a(x)f(x) + b(x)P(x) = 1$ for some $a(x), b(x) \in F[x]$,

$a(\alpha)\cdot f(\alpha) + b(\alpha)\cdot P(\alpha) = a(\alpha)\cdot f(\alpha) + 0 = 1$, or

$$f(\alpha)^{-1} = a(\alpha).$$